Aryan Pratap Singh

Software Engineer — Full-Stack, Backend & Generative AI Systems

+91 7007839057 | aryansingh7598@gmail.com | linkedin.com/in/aryan64 | github.com/AryanSingh64 | aryans-portfolio-pi.vercel.app

Summary

Product-minded software engineer with hands-on experience building, scaling, and deploying full-stack web platforms, asynchronous backends, and Generative AI systems. Proficient in Python, TypeScript, FastAPI, React/Next.js 15, PostgreSQL, and MongoDB. Built and launched 3 live production applications, including an offline multimodal RAG platform and an autonomous code patch agent. National Semifinalist at Rift Hackathon (700+ teams). High-agency builder comfortable with system design (HLD/LLD), rapid experimentation, and taking features from conception to production. Class of 2027.

Education

Pranveer Singh Institute of Technology — Kanpur, India
Bachelor of Technology in Information Technology — *Oct. 2023 – May 2027*

Technical Skills

Languages: Python, TypeScript, JavaScript (ES6+), Java, SQL, C++

Backend & APIs: FastAPI (Asyncio), Node.js, NestJS, Express.js, REST APIs, WebSockets, Server-Sent Events (SSE), OAuth 2.0, JWT

Frontend Development: React, Next.js 15, Tailwind CSS, D3.js, HTML5, CSS3, Responsive UI Architecture

AI & GenAI Systems: RAG Pipelines, Semantic & Hybrid Search, Vector DBs (ChromaDB), Local LLMs (Ollama), OpenAI/Gemini/Groq APIs

Databases & DevOps: PostgreSQL, MongoDB, Supabase, Docker, Linux, Git, GitHub Actions (CI/CD), Vercel, Redis Concepts

Core Engineering: System Design (HLD/LLD), Data Structures & Algorithms, OOP, TDD (pytest, Playwright), SDLC, Agile

Featured Production Projects

Mi:RAG Live: mirag.me | GitHub — *Aug 2026*

Offline multimodal document intelligence platform with semantic vector retrieval local serving

Async Python, FastAPI, ChromaDB, Sentence-Transformers, PyMuPDF, RapidOCR, Ollama, Docker

- Architected an end-to-end document intelligence platform extracting text, formulas, and diagrams from complex PDFs using PyMuPDF drawing-path parsing and RapidOCR, generating 384-dimensional vector embeddings stored in ChromaDB.
- Engineered an asynchronous FastAPI REST backend supporting multi-turn semantic search with dynamic query rewriting and local Ollama LLM inference constrained by strict anti-hallucination guardrails.
- Authored hardware detection scripts and an automated packaging pipeline that bundles model weights, vector stores, and runtimes into a zero-configuration, self-contained offline distribution.

GitFixAI Live: gitfixai.vercel.app | GitHub — *Feb 2026*

Full-stack autonomous developer platform for automated issue triage and code patch generation

React, TypeScript, FastAPI, WebSockets, Supabase (PostgreSQL), GitHub API, OpenAI API

- Built a full-stack developer tool with a React/TypeScript UI and FastAPI backend that parses GitHub issues, performs repository-level code reasoning via LLM APIs, and outputs verified patch diffs in under 45 seconds.
- Engineered bidirectional WebSocket streaming for live execution logs, integrating GitHub OAuth 2.0 to automate feature branch provisioning, code commits, and pull request deployment.
- Implemented automated test suites with pytest covering API validation, token refresh edge cases, and rate-limiting fallbacks.

ThreadLine Live: thread-line-five.vercel.app | GitHub — *Apr 2026*

High-concurrency data aggregation and interactive visualization platform

Next.js 15, TypeScript, React, D3.js, Server-Sent Events, Playwright, TailwindCSS, Groq API

- Architected an asynchronous pipeline querying 150+ external endpoints concurrently, using Server-Sent Events (SSE) to stream live scan results to a Next.js 15 frontend without blocking the UI thread.
- Developed dynamic risk scoring heuristics, automated summary synthesis using the Groq API (Llama 3), and an interactive D3.js force-directed entity relationship graph. Automated end-to-end testing with Playwright.

Achievements & Leadership

- **National Semifinalist, Rift Hackathon 2026 (Physics Wallah Institute of Innovation):** Advanced to national semifinals with GitFixAI, competing among 700+ teams across 4 cities on product engineering and execution.
- **Top 5%, IntrusionX Hackathon (GLA University, 2026):** Ranked 17th out of 354 participating teams building full-stack developer tooling under strict hackathon deadlines.
- **Competitive Programming:** Solved 150+ Data Structures and Algorithms problems across LeetCode and GeeksforGeeks.
- **Event Coordinator, Inter-College Chess Tournament (College Club):** Led end-to-end event operations, scheduling, and match logistics for a 600+ participant multi-stage tournament.

Certifications

- **DeepLearning.AI:** Retrieval Augmented Generation (RAG) · Agentic AI · Generative AI for Software Development